Table 15 WHO base anthropometric outcomes for primary school, oldest child

Current school enrollment defines the primary school sample. One oldest child per household is selected within this level. WHO 2007 sex-specific monthly growth references use age on 16 December 2022. WAZ is defined only through month 120. Father and mother reform estimates use the established weighted RDD.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.256	0.084 [0.196] N=1,974	0.183 [0.205] N=1,822	0.329 [0.210] N=1,641	0.303 [0.227] N=1,434	**0.465* [0.239] N=1,200**
Weight-for-age z score, through 120 months	0.322	0.010 [0.137] N=1,967	0.028 [0.147] N=1,815	0.127 [0.155] N=1,635	0.154 [0.162] N=1,424	0.202 [0.171] N=1,192
BMI-for-age z score	0.575	-0.042 [0.195] N=1,942	-0.061 [0.209] N=1,789	-0.003 [0.219] N=1,612	-0.003 [0.237] N=1,408	0.045 [0.258] N=1,179
Stunted, HAZ below -2	0.171 P=328	-0.043 [0.038] N=1,974	-0.059 [0.039] N=1,822	**-0.072* [0.042] N=1,641**	**-0.078* [0.045] N=1,434**	**-0.118** [0.050] N=1,200**
Underweight, WAZ below -2, through 120 months	0.045 P=88	0.022 [0.020] N=1,967	0.028 [0.020] N=1,815	0.021 [0.022] N=1,635	0.035 [0.023] N=1,424	0.039 [0.025] N=1,192
Thinness, BMIZ below -2	0.081 P=156	0.026 [0.026] N=1,942	0.021 [0.028] N=1,789	0.020 [0.030] N=1,612	**0.051* [0.028] N=1,408**	0.039 [0.032] N=1,179
Overweight including obesity, BMIZ above +1	0.412 P=800	0.037 [0.051] N=1,942	0.017 [0.055] N=1,789	0.029 [0.058] N=1,612	0.057 [0.064] N=1,408	0.052 [0.068] N=1,179
Obesity, BMIZ above +2	0.217 P=416	**0.070* [0.042] N=1,942**	0.057 [0.044] N=1,789	**0.083* [0.048] N=1,612**	**0.093* [0.049] N=1,408**	0.068 [0.053] N=1,179

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.385 [0.290] N=1,666	-0.306 [0.321] N=1,514	-0.129 [0.355] N=1,333	-0.358 [0.397] N=1,126	-0.294 [0.489] N=892
Weight-for-age z score, through 120 months	-0.155 [0.175] N=1,661	-0.166 [0.202] N=1,509	-0.022 [0.223] N=1,329	-0.062 [0.249] N=1,118	-0.099 [0.290] N=886
BMI-for-age z score	0.039 [0.266] N=1,640	0.009 [0.299] N=1,487	0.135 [0.327] N=1,310	0.132 [0.378] N=1,106	0.202 [0.471] N=877
Stunted, HAZ below -2	0.051 [0.053] N=1,666	0.044 [0.057] N=1,514	0.040 [0.062] N=1,333	0.061 [0.069] N=1,126	0.025 [0.087] N=892
Underweight, WAZ below -2, through 120 months	0.022 [0.028] N=1,661	0.034 [0.030] N=1,509	0.017 [0.034] N=1,329	0.052 [0.036] N=1,118	**0.076** [0.035] N=886**
Thinness, BMIZ below -2	-0.007 [0.036] N=1,640	-0.024 [0.041] N=1,487	-0.037 [0.044] N=1,310	0.017 [0.045] N=1,106	-0.005 [0.061] N=877
Overweight including obesity, BMIZ above +1	0.048 [0.078] N=1,640	0.009 [0.089] N=1,487	0.032 [0.100] N=1,310	0.088 [0.120] N=1,106	0.071 [0.147] N=877
Obesity, BMIZ above +2	0.080 [0.056] N=1,640	0.057 [0.062] N=1,487	0.106 [0.072] N=1,310	**0.136* [0.078] N=1,106**	0.083 [0.096] N=877

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.256	0.076 [0.193] N=1,974	0.176 [0.202] N=1,822	0.322 [0.208] N=1,641	0.297 [0.224] N=1,434	**0.446* [0.234] N=1,200**
Weight-for-age z score, through 120 months	0.322	-0.002 [0.134] N=1,967	0.014 [0.144] N=1,815	0.109 [0.152] N=1,635	0.132 [0.158] N=1,424	0.169 [0.166] N=1,192
BMI-for-age z score	0.575	-0.047 [0.196] N=1,942	-0.069 [0.209] N=1,789	-0.013 [0.220] N=1,612	-0.017 [0.238] N=1,408	0.029 [0.259] N=1,179
Stunted, HAZ below -2	0.171 P=328	-0.043 [0.038] N=1,974	-0.059 [0.039] N=1,822	**-0.071* [0.042] N=1,641**	**-0.079* [0.045] N=1,434**	**-0.117** [0.051] N=1,200**
Underweight, WAZ below -2, through 120 months	0.045 P=88	0.023 [0.020] N=1,967	0.028 [0.020] N=1,815	0.021 [0.022] N=1,635	0.036 [0.023] N=1,424	0.040 [0.024] N=1,192
Thinness, BMIZ below -2	0.081 P=156	0.025 [0.026] N=1,942	0.020 [0.028] N=1,789	0.019 [0.030] N=1,612	**0.050* [0.028] N=1,408**	0.037 [0.031] N=1,179
Overweight including obesity, BMIZ above +1	0.412 P=800	0.035 [0.052] N=1,942	0.014 [0.056] N=1,789	0.026 [0.059] N=1,612	0.052 [0.065] N=1,408	0.047 [0.069] N=1,179
Obesity, BMIZ above +2	0.217 P=416	0.068 [0.042] N=1,942	0.054 [0.045] N=1,789	**0.080* [0.048] N=1,612**	**0.089* [0.050] N=1,408**	0.066 [0.054] N=1,179

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.383 [0.290] N=1,666	-0.307 [0.322] N=1,514	-0.131 [0.355] N=1,333	-0.362 [0.398] N=1,126	-0.295 [0.490] N=892
Weight-for-age z score, through 120 months	-0.141 [0.176] N=1,661	-0.156 [0.202] N=1,509	-0.009 [0.223] N=1,329	-0.053 [0.248] N=1,118	-0.091 [0.288] N=886
BMI-for-age z score	0.055 [0.269] N=1,640	0.022 [0.302] N=1,487	0.152 [0.330] N=1,310	0.145 [0.381] N=1,106	0.217 [0.477] N=877
Stunted, HAZ below -2	0.051 [0.053] N=1,666	0.044 [0.057] N=1,514	0.041 [0.062] N=1,333	0.062 [0.069] N=1,126	0.025 [0.087] N=892
Underweight, WAZ below -2, through 120 months	0.022 [0.029] N=1,661	0.034 [0.030] N=1,509	0.017 [0.034] N=1,329	0.052 [0.036] N=1,118	**0.076** [0.035] N=886**
Thinness, BMIZ below -2	-0.004 [0.036] N=1,640	-0.023 [0.041] N=1,487	-0.036 [0.044] N=1,310	0.018 [0.045] N=1,106	-0.004 [0.061] N=877
Overweight including obesity, BMIZ above +1	0.054 [0.079] N=1,640	0.015 [0.090] N=1,487	0.039 [0.102] N=1,310	0.092 [0.122] N=1,106	0.077 [0.150] N=877
Obesity, BMIZ above +2	0.085 [0.057] N=1,640	0.062 [0.063] N=1,487	0.111 [0.073] N=1,310	**0.139* [0.079] N=1,106**	0.086 [0.097] N=877

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.256	0.062 [0.190] N=1,974	0.147 [0.199] N=1,822	0.283 [0.205] N=1,641	0.227 [0.222] N=1,434	0.366 [0.234] N=1,200
Weight-for-age z score, through 120 months	0.322	-0.012 [0.133] N=1,967	-0.003 [0.145] N=1,815	0.087 [0.152] N=1,635	0.105 [0.157] N=1,424	0.139 [0.167] N=1,192
BMI-for-age z score	0.575	-0.050 [0.196] N=1,942	-0.070 [0.209] N=1,789	-0.010 [0.219] N=1,612	0.002 [0.235] N=1,408	0.059 [0.256] N=1,179
Stunted, HAZ below -2	0.171 P=328	-0.041 [0.038] N=1,974	-0.056 [0.039] N=1,822	-0.067 [0.042] N=1,641	-0.071 [0.045] N=1,434	**-0.106** [0.051] N=1,200**
Underweight, WAZ below -2, through 120 months	0.045 P=88	0.024 [0.020] N=1,967	0.029 [0.020] N=1,815	0.022 [0.022] N=1,635	**0.039* [0.023] N=1,424**	**0.042* [0.024] N=1,192**
Thinness, BMIZ below -2	0.081 P=156	0.026 [0.026] N=1,942	0.019 [0.028] N=1,789	0.017 [0.030] N=1,612	0.046 [0.028] N=1,408	0.033 [0.031] N=1,179
Overweight including obesity, BMIZ above +1	0.412 P=800	0.036 [0.052] N=1,942	0.015 [0.055] N=1,789	0.027 [0.059] N=1,612	0.058 [0.064] N=1,408	0.056 [0.068] N=1,179
Obesity, BMIZ above +2	0.217 P=416	0.066 [0.043] N=1,942	0.052 [0.045] N=1,789	0.077 [0.048] N=1,612	**0.088* [0.050] N=1,408**	0.066 [0.054] N=1,179

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.412 [0.281] N=1,666	-0.369 [0.309] N=1,514	-0.218 [0.341] N=1,333	-0.528 [0.375] N=1,126	-0.525 [0.456] N=892
Weight-for-age z score, through 120 months	-0.160 [0.175] N=1,661	-0.191 [0.202] N=1,509	-0.058 [0.222] N=1,329	-0.119 [0.248] N=1,118	-0.183 [0.286] N=886
BMI-for-age z score	0.051 [0.269] N=1,640	0.022 [0.302] N=1,487	0.158 [0.328] N=1,310	0.184 [0.378] N=1,106	0.291 [0.468] N=877
Stunted, HAZ below -2	0.057 [0.051] N=1,666	0.053 [0.055] N=1,514	0.054 [0.060] N=1,333	0.085 [0.066] N=1,126	0.056 [0.081] N=892
Underweight, WAZ below -2, through 120 months	0.023 [0.029] N=1,661	0.036 [0.030] N=1,509	0.020 [0.035] N=1,329	0.059 [0.037] N=1,118	**0.084** [0.036] N=886**
Thinness, BMIZ below -2	-0.004 [0.036] N=1,640	-0.024 [0.041] N=1,487	-0.040 [0.044] N=1,310	0.011 [0.044] N=1,106	-0.015 [0.060] N=877
Overweight including obesity, BMIZ above +1	0.053 [0.079] N=1,640	0.014 [0.091] N=1,487	0.037 [0.102] N=1,310	0.093 [0.122] N=1,106	0.085 [0.150] N=877
Obesity, BMIZ above +2	0.082 [0.058] N=1,640	0.058 [0.064] N=1,487	0.106 [0.073] N=1,310	**0.140* [0.079] N=1,106**	0.092 [0.099] N=877

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.256	0.045 [0.191] N=1,974	0.116 [0.201] N=1,822	0.252 [0.207] N=1,641	0.212 [0.221] N=1,434	0.331 [0.234] N=1,200
Weight-for-age z score, through 120 months	0.322	-0.022 [0.132] N=1,967	-0.020 [0.144] N=1,815	0.068 [0.152] N=1,635	0.086 [0.157] N=1,424	0.113 [0.168] N=1,192
BMI-for-age z score	0.575	-0.047 [0.196] N=1,942	-0.065 [0.209] N=1,789	-0.010 [0.221] N=1,612	-0.011 [0.237] N=1,408	0.052 [0.258] N=1,179
Stunted, HAZ below -2	0.171 P=328	-0.036 [0.038] N=1,974	-0.048 [0.040] N=1,822	-0.059 [0.042] N=1,641	-0.064 [0.045] N=1,434	**-0.095* [0.051] N=1,200**
Underweight, WAZ below -2, through 120 months	0.045 P=88	0.024 [0.020] N=1,967	0.031 [0.020] N=1,815	0.025 [0.022] N=1,635	**0.041* [0.023] N=1,424**	**0.045* [0.025] N=1,192**
Thinness, BMIZ below -2	0.081 P=156	0.023 [0.026] N=1,942	0.016 [0.028] N=1,789	0.016 [0.030] N=1,612	0.046 [0.029] N=1,408	0.029 [0.031] N=1,179
Overweight including obesity, BMIZ above +1	0.412 P=800	0.035 [0.052] N=1,942	0.014 [0.056] N=1,789	0.026 [0.059] N=1,612	0.055 [0.064] N=1,408	0.053 [0.069] N=1,179
Obesity, BMIZ above +2	0.217 P=416	0.066 [0.043] N=1,942	0.052 [0.045] N=1,789	0.078 [0.048] N=1,612	**0.089* [0.050] N=1,408**	0.068 [0.054] N=1,179

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.394 [0.287] N=1,666	-0.374 [0.317] N=1,514	-0.222 [0.348] N=1,333	-0.493 [0.383] N=1,126	-0.521 [0.481] N=892
Weight-for-age z score, through 120 months	-0.152 [0.171] N=1,661	-0.195 [0.198] N=1,509	-0.062 [0.217] N=1,329	-0.117 [0.238] N=1,118	-0.200 [0.278] N=886
BMI-for-age z score	0.045 [0.268] N=1,640	0.017 [0.302] N=1,487	0.152 [0.327] N=1,310	0.154 [0.375] N=1,106	0.252 [0.469] N=877
Stunted, HAZ below -2	0.055 [0.052] N=1,666	0.056 [0.056] N=1,514	0.055 [0.061] N=1,333	0.079 [0.066] N=1,126	0.057 [0.082] N=892
Underweight, WAZ below -2, through 120 months	0.021 [0.028] N=1,661	0.035 [0.030] N=1,509	0.020 [0.034] N=1,329	0.056 [0.036] N=1,118	**0.084** [0.036] N=886**
Thinness, BMIZ below -2	-0.006 [0.036] N=1,640	-0.026 [0.042] N=1,487	-0.039 [0.044] N=1,310	0.014 [0.046] N=1,106	-0.015 [0.061] N=877
Overweight including obesity, BMIZ above +1	0.052 [0.079] N=1,640	0.011 [0.090] N=1,487	0.035 [0.100] N=1,310	0.089 [0.119] N=1,106	0.081 [0.145] N=877
Obesity, BMIZ above +2	0.080 [0.058] N=1,640	0.055 [0.064] N=1,487	0.101 [0.073] N=1,310	**0.130* [0.077] N=1,106**	0.081 [0.096] N=877

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.267	0.096 [0.208] N=1,858	0.203 [0.218] N=1,715	0.341 [0.227] N=1,542	0.319 [0.244] N=1,349	**0.486* [0.258] N=1,127**
Weight-for-age z score, through 120 months	0.323	-0.070 [0.142] N=1,852	-0.061 [0.152] N=1,709	0.036 [0.162] N=1,537	0.060 [0.169] N=1,340	0.122 [0.182] N=1,119
BMI-for-age z score	0.586	-0.117 [0.200] N=1,830	-0.150 [0.212] N=1,686	-0.088 [0.222] N=1,517	-0.107 [0.241] N=1,327	-0.082 [0.264] N=1,108
Stunted, HAZ below -2	0.174 P=314	-0.045 [0.040] N=1,858	-0.059 [0.041] N=1,715	-0.069 [0.044] N=1,542	-0.078 [0.048] N=1,349	**-0.118** [0.053] N=1,127**
Underweight, WAZ below -2, through 120 months	0.045 P=83	0.028 [0.021] N=1,852	0.034 [0.021] N=1,709	0.028 [0.024] N=1,537	**0.045* [0.024] N=1,340**	0.041 [0.026] N=1,119
Thinness, BMIZ below -2	0.081 P=148	0.025 [0.027] N=1,830	0.020 [0.029] N=1,686	0.019 [0.031] N=1,517	**0.049* [0.030] N=1,327**	0.045 [0.033] N=1,108
Overweight including obesity, BMIZ above +1	0.416 P=761	0.019 [0.053] N=1,830	-0.003 [0.056] N=1,686	0.013 [0.059] N=1,517	0.036 [0.065] N=1,327	0.024 [0.070] N=1,108
Obesity, BMIZ above +2	0.217 P=393	0.048 [0.042] N=1,830	0.034 [0.044] N=1,686	0.067 [0.047] N=1,517	0.074 [0.049] N=1,327	0.050 [0.053] N=1,108

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.347 [0.294] N=1,574	-0.261 [0.323] N=1,431	-0.106 [0.359] N=1,258	-0.353 [0.401] N=1,065	-0.252 [0.515] N=843
Weight-for-age z score, through 120 months	-0.187 [0.175] N=1,569	-0.214 [0.199] N=1,426	-0.064 [0.221] N=1,254	-0.104 [0.244] N=1,057	-0.074 [0.296] N=836
BMI-for-age z score	0.001 [0.276] N=1,551	-0.041 [0.308] N=1,407	0.112 [0.334] N=1,238	0.104 [0.384] N=1,048	0.156 [0.498] N=829
Stunted, HAZ below -2	0.041 [0.053] N=1,574	0.037 [0.057] N=1,431	0.040 [0.064] N=1,258	0.064 [0.071] N=1,065	0.021 [0.088] N=843
Underweight, WAZ below -2, through 120 months	0.025 [0.030] N=1,569	0.038 [0.031] N=1,426	0.026 [0.035] N=1,254	**0.064* [0.037] N=1,057**	**0.067* [0.038] N=836**
Thinness, BMIZ below -2	-0.002 [0.038] N=1,551	-0.020 [0.043] N=1,407	-0.034 [0.046] N=1,238	0.020 [0.047] N=1,048	0.018 [0.065] N=829
Overweight including obesity, BMIZ above +1	0.039 [0.081] N=1,551	-0.003 [0.092] N=1,407	0.029 [0.103] N=1,238	0.080 [0.124] N=1,048	0.067 [0.155] N=829
Obesity, BMIZ above +2	0.065 [0.057] N=1,551	0.041 [0.063] N=1,407	0.107 [0.073] N=1,238	**0.135* [0.078] N=1,048**	0.090 [0.099] N=829

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.267	0.084 [0.208] N=1,858	0.200 [0.217] N=1,715	0.344 [0.225] N=1,542	0.321 [0.242] N=1,349	**0.509** [0.254] N=1,127**
Weight-for-age z score, through 120 months	0.323	-0.078 [0.141] N=1,852	-0.065 [0.152] N=1,709	0.034 [0.161] N=1,537	0.062 [0.169] N=1,340	0.130 [0.181] N=1,119
BMI-for-age z score	0.586	-0.108 [0.199] N=1,830	-0.145 [0.212] N=1,686	-0.084 [0.222] N=1,517	-0.100 [0.240] N=1,327	-0.074 [0.264] N=1,108
Stunted, HAZ below -2	0.174 P=314	-0.043 [0.039] N=1,858	-0.059 [0.040] N=1,715	-0.070 [0.043] N=1,542	**-0.080* [0.047] N=1,349**	**-0.124** [0.053] N=1,127**
Underweight, WAZ below -2, through 120 months	0.045 P=83	0.028 [0.021] N=1,852	0.034 [0.021] N=1,709	0.028 [0.024] N=1,537	**0.045* [0.025] N=1,340**	0.041 [0.026] N=1,119
Thinness, BMIZ below -2	0.081 P=148	0.027 [0.027] N=1,830	0.021 [0.029] N=1,686	0.020 [0.031] N=1,517	**0.050* [0.030] N=1,327**	0.044 [0.034] N=1,108
Overweight including obesity, BMIZ above +1	0.416 P=761	0.020 [0.053] N=1,830	-0.003 [0.057] N=1,686	0.013 [0.060] N=1,517	0.037 [0.065] N=1,327	0.024 [0.070] N=1,108
Obesity, BMIZ above +2	0.217 P=393	0.052 [0.042] N=1,830	0.036 [0.044] N=1,686	0.069 [0.047] N=1,517	0.076 [0.049] N=1,327	0.051 [0.052] N=1,108

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.362 [0.295] N=1,574	-0.263 [0.323] N=1,431	-0.097 [0.358] N=1,258	-0.341 [0.399] N=1,065	-0.180 [0.506] N=843
Weight-for-age z score, through 120 months	-0.194 [0.174] N=1,569	-0.214 [0.199] N=1,426	-0.061 [0.222] N=1,254	-0.090 [0.247] N=1,057	-0.049 [0.302] N=836
BMI-for-age z score	0.018 [0.275] N=1,551	-0.033 [0.308] N=1,407	0.120 [0.335] N=1,238	0.122 [0.386] N=1,048	0.166 [0.501] N=829
Stunted, HAZ below -2	0.044 [0.053] N=1,574	0.037 [0.057] N=1,431	0.038 [0.063] N=1,258	0.060 [0.071] N=1,065	0.003 [0.088] N=843
Underweight, WAZ below -2, through 120 months	0.025 [0.030] N=1,569	0.038 [0.031] N=1,426	0.025 [0.036] N=1,254	0.063 [0.038] N=1,057	**0.067* [0.038] N=836**
Thinness, BMIZ below -2	-0.000 [0.038] N=1,551	-0.018 [0.043] N=1,407	-0.033 [0.046] N=1,238	0.021 [0.047] N=1,048	0.015 [0.064] N=829
Overweight including obesity, BMIZ above +1	0.041 [0.082] N=1,551	-0.001 [0.093] N=1,407	0.033 [0.105] N=1,238	0.088 [0.126] N=1,048	0.069 [0.158] N=829
Obesity, BMIZ above +2	0.069 [0.057] N=1,551	0.045 [0.063] N=1,407	0.109 [0.073] N=1,238	**0.139* [0.078] N=1,048**	0.086 [0.099] N=829

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.138	0.048 [0.139] N=2,411	0.122 [0.148] N=2,200	0.103 [0.154] N=1,950	0.016 [0.167] N=1,678	0.009 [0.192] N=1,357
Weight-for-age z score, through 120 months	0.398	0.021 [0.103] N=2,378	0.059 [0.107] N=2,177	0.107 [0.116] N=1,933	0.090 [0.118] N=1,660	0.037 [0.128] N=1,344
BMI-for-age z score	0.579	0.170 [0.143] N=2,375	0.184 [0.151] N=2,169	0.246 [0.164] N=1,922	0.248 [0.174] N=1,650	0.189 [0.195] N=1,331
Stunted, HAZ below -2	0.157 P=366	-0.001 [0.029] N=2,411	-0.019 [0.032] N=2,200	-0.003 [0.034] N=1,950	0.008 [0.037] N=1,678	-0.006 [0.041] N=1,357
Underweight, WAZ below -2, through 120 months	0.039 P=92	0.009 [0.014] N=2,378	0.003 [0.015] N=2,177	-0.004 [0.015] N=1,933	0.002 [0.017] N=1,660	0.010 [0.019] N=1,344
Thinness, BMIZ below -2	0.081 P=193	-0.011 [0.020] N=2,375	-0.012 [0.020] N=2,169	-0.006 [0.021] N=1,922	-0.015 [0.023] N=1,650	-0.012 [0.025] N=1,331
Overweight including obesity, BMIZ above +1	0.417 P=982	0.017 [0.042] N=2,375	0.011 [0.044] N=2,169	0.031 [0.048] N=1,922	0.011 [0.052] N=1,650	-0.016 [0.058] N=1,331
Obesity, BMIZ above +2	0.214 P=499	0.012 [0.034] N=2,375	0.017 [0.035] N=2,169	0.041 [0.039] N=1,922	0.037 [0.043] N=1,650	-0.000 [0.048] N=1,331

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.078 [0.181] N=2,046	0.227 [0.195] N=1,835	0.220 [0.209] N=1,585	0.045 [0.241] N=1,313	0.113 [0.305] N=992
Weight-for-age z score, through 120 months	-0.056 [0.164] N=2,014	0.000 [0.180] N=1,813	0.092 [0.211] N=1,569	0.038 [0.241] N=1,296	-0.146 [0.289] N=980
BMI-for-age z score	0.103 [0.206] N=2,014	0.113 [0.227] N=1,808	0.230 [0.266] N=1,561	0.225 [0.313] N=1,289	-0.019 [0.396] N=970
Stunted, HAZ below -2	0.000 [0.039] N=2,046	-0.030 [0.044] N=1,835	-0.007 [0.048] N=1,585	0.022 [0.058] N=1,313	-0.024 [0.068] N=992
Underweight, WAZ below -2, through 120 months	0.018 [0.021] N=2,014	0.008 [0.022] N=1,813	-0.008 [0.024] N=1,569	-0.004 [0.029] N=1,296	0.013 [0.040] N=980
Thinness, BMIZ below -2	-0.004 [0.031] N=2,014	-0.007 [0.032] N=1,808	0.007 [0.038] N=1,561	-0.013 [0.041] N=1,289	-0.004 [0.055] N=970
Overweight including obesity, BMIZ above +1	0.033 [0.061] N=2,014	0.023 [0.066] N=1,808	0.068 [0.076] N=1,561	0.026 [0.092] N=1,289	-0.050 [0.114] N=970
Obesity, BMIZ above +2	-0.016 [0.047] N=2,014	-0.010 [0.052] N=1,808	0.030 [0.061] N=1,561	0.014 [0.076] N=1,289	-0.135 [0.092] N=970

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.138	0.050 [0.138] N=2,411	0.124 [0.147] N=2,200	0.103 [0.154] N=1,950	0.015 [0.167] N=1,678	0.007 [0.192] N=1,357
Weight-for-age z score, through 120 months	0.398	0.028 [0.103] N=2,378	0.064 [0.107] N=2,177	0.110 [0.116] N=1,933	0.085 [0.118] N=1,660	0.032 [0.127] N=1,344
BMI-for-age z score	0.579	0.173 [0.144] N=2,375	0.185 [0.151] N=2,169	0.245 [0.165] N=1,922	0.241 [0.176] N=1,650	0.183 [0.196] N=1,331
Stunted, HAZ below -2	0.157 P=366	-0.001 [0.030] N=2,411	-0.019 [0.032] N=2,200	-0.003 [0.034] N=1,950	0.008 [0.037] N=1,678	-0.006 [0.041] N=1,357
Underweight, WAZ below -2, through 120 months	0.039 P=92	0.009 [0.014] N=2,378	0.003 [0.015] N=2,177	-0.004 [0.015] N=1,933	0.003 [0.016] N=1,660	0.010 [0.019] N=1,344
Thinness, BMIZ below -2	0.081 P=193	-0.011 [0.020] N=2,375	-0.012 [0.020] N=2,169	-0.006 [0.021] N=1,922	-0.015 [0.022] N=1,650	-0.011 [0.025] N=1,331
Overweight including obesity, BMIZ above +1	0.417 P=982	0.018 [0.043] N=2,375	0.012 [0.045] N=2,169	0.031 [0.049] N=1,922	0.009 [0.052] N=1,650	-0.018 [0.058] N=1,331
Obesity, BMIZ above +2	0.214 P=499	0.013 [0.034] N=2,375	0.017 [0.035] N=2,169	0.041 [0.039] N=1,922	0.035 [0.043] N=1,650	-0.002 [0.048] N=1,331

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.085 [0.180] N=2,046	0.234 [0.195] N=1,835	0.223 [0.208] N=1,585	0.047 [0.240] N=1,313	0.108 [0.305] N=992
Weight-for-age z score, through 120 months	-0.031 [0.164] N=2,014	0.031 [0.178] N=1,813	0.122 [0.210] N=1,569	0.064 [0.240] N=1,296	-0.103 [0.287] N=980
BMI-for-age z score	0.122 [0.206] N=2,014	0.134 [0.228] N=1,808	0.256 [0.266] N=1,561	0.249 [0.315] N=1,289	0.026 [0.395] N=970
Stunted, HAZ below -2	0.002 [0.039] N=2,046	-0.029 [0.044] N=1,835	-0.006 [0.048] N=1,585	0.025 [0.059] N=1,313	-0.019 [0.068] N=992
Underweight, WAZ below -2, through 120 months	0.018 [0.021] N=2,014	0.008 [0.022] N=1,813	-0.008 [0.024] N=1,569	-0.005 [0.029] N=1,296	0.012 [0.041] N=980
Thinness, BMIZ below -2	-0.002 [0.030] N=2,014	-0.005 [0.032] N=1,808	0.007 [0.038] N=1,561	-0.013 [0.042] N=1,289	-0.005 [0.056] N=970
Overweight including obesity, BMIZ above +1	0.041 [0.061] N=2,014	0.031 [0.066] N=1,808	0.076 [0.076] N=1,561	0.034 [0.092] N=1,289	-0.036 [0.114] N=970
Obesity, BMIZ above +2	-0.008 [0.047] N=2,014	-0.002 [0.051] N=1,808	0.039 [0.061] N=1,561	0.023 [0.076] N=1,289	-0.120 [0.092] N=970

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.138	0.054 [0.134] N=2,411	0.119 [0.144] N=2,200	0.094 [0.150] N=1,950	0.007 [0.162] N=1,678	0.004 [0.187] N=1,357
Weight-for-age z score, through 120 months	0.398	0.036 [0.101] N=2,378	0.066 [0.105] N=2,177	0.103 [0.114] N=1,933	0.080 [0.115] N=1,660	0.024 [0.125] N=1,344
BMI-for-age z score	0.579	0.175 [0.144] N=2,375	0.188 [0.151] N=2,169	0.246 [0.165] N=1,922	0.243 [0.176] N=1,650	0.182 [0.197] N=1,331
Stunted, HAZ below -2	0.157 P=366	-0.002 [0.029] N=2,411	-0.018 [0.031] N=2,200	-0.002 [0.033] N=1,950	0.009 [0.036] N=1,678	-0.006 [0.039] N=1,357
Underweight, WAZ below -2, through 120 months	0.039 P=92	0.009 [0.014] N=2,378	0.003 [0.015] N=2,177	-0.003 [0.015] N=1,933	0.003 [0.016] N=1,660	0.011 [0.019] N=1,344
Thinness, BMIZ below -2	0.081 P=193	-0.011 [0.020] N=2,375	-0.012 [0.020] N=2,169	-0.006 [0.022] N=1,922	-0.015 [0.023] N=1,650	-0.011 [0.025] N=1,331
Overweight including obesity, BMIZ above +1	0.417 P=982	0.019 [0.043] N=2,375	0.012 [0.045] N=2,169	0.030 [0.049] N=1,922	0.008 [0.052] N=1,650	-0.021 [0.058] N=1,331
Obesity, BMIZ above +2	0.214 P=499	0.015 [0.034] N=2,375	0.020 [0.036] N=2,169	0.044 [0.039] N=1,922	0.038 [0.043] N=1,650	-0.001 [0.048] N=1,331

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.092 [0.181] N=2,046	0.229 [0.196] N=1,835	0.203 [0.209] N=1,585	0.026 [0.236] N=1,313	0.084 [0.304] N=992
Weight-for-age z score, through 120 months	-0.018 [0.161] N=2,014	0.035 [0.176] N=1,813	0.110 [0.206] N=1,569	0.050 [0.233] N=1,296	-0.131 [0.280] N=980
BMI-for-age z score	0.124 [0.207] N=2,014	0.136 [0.228] N=1,808	0.257 [0.267] N=1,561	0.249 [0.312] N=1,289	0.022 [0.394] N=970
Stunted, HAZ below -2	0.001 [0.039] N=2,046	-0.029 [0.044] N=1,835	-0.003 [0.048] N=1,585	0.028 [0.058] N=1,313	-0.018 [0.068] N=992
Underweight, WAZ below -2, through 120 months	0.017 [0.021] N=2,014	0.007 [0.022] N=1,813	-0.008 [0.025] N=1,569	-0.004 [0.029] N=1,296	0.014 [0.043] N=980
Thinness, BMIZ below -2	-0.002 [0.030] N=2,014	-0.006 [0.032] N=1,808	0.005 [0.038] N=1,561	-0.014 [0.041] N=1,289	-0.008 [0.054] N=970
Overweight including obesity, BMIZ above +1	0.042 [0.061] N=2,014	0.031 [0.066] N=1,808	0.074 [0.076] N=1,561	0.032 [0.091] N=1,289	-0.042 [0.113] N=970
Obesity, BMIZ above +2	-0.006 [0.046] N=2,014	-0.000 [0.051] N=1,808	0.040 [0.060] N=1,561	0.025 [0.075] N=1,289	-0.122 [0.092] N=970

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.138	0.036 [0.132] N=2,411	0.100 [0.141] N=2,200	0.067 [0.147] N=1,950	-0.011 [0.160] N=1,678	-0.045 [0.188] N=1,357
Weight-for-age z score, through 120 months	0.398	0.025 [0.098] N=2,378	0.058 [0.102] N=2,177	0.096 [0.110] N=1,933	0.076 [0.111] N=1,660	0.006 [0.119] N=1,344
BMI-for-age z score	0.579	0.174 [0.144] N=2,375	0.189 [0.151] N=2,169	0.251 [0.165] N=1,922	0.250 [0.176] N=1,650	0.190 [0.196] N=1,331
Stunted, HAZ below -2	0.157 P=366	0.002 [0.028] N=2,411	-0.014 [0.031] N=2,200	0.003 [0.032] N=1,950	0.012 [0.035] N=1,678	0.003 [0.040] N=1,357
Underweight, WAZ below -2, through 120 months	0.039 P=92	0.009 [0.014] N=2,378	0.004 [0.015] N=2,177	-0.003 [0.015] N=1,933	0.003 [0.016] N=1,660	0.012 [0.018] N=1,344
Thinness, BMIZ below -2	0.081 P=193	-0.011 [0.020] N=2,375	-0.012 [0.020] N=2,169	-0.006 [0.021] N=1,922	-0.016 [0.022] N=1,650	-0.011 [0.024] N=1,331
Overweight including obesity, BMIZ above +1	0.417 P=982	0.019 [0.043] N=2,375	0.013 [0.045] N=2,169	0.032 [0.049] N=1,922	0.010 [0.052] N=1,650	-0.016 [0.058] N=1,331
Obesity, BMIZ above +2	0.214 P=499	0.014 [0.034] N=2,375	0.018 [0.035] N=2,169	0.042 [0.038] N=1,922	0.035 [0.042] N=1,650	-0.001 [0.048] N=1,331

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.095 [0.180] N=2,046	0.237 [0.196] N=1,835	0.220 [0.210] N=1,585	0.062 [0.242] N=1,313	0.079 [0.314] N=992
Weight-for-age z score, through 120 months	-0.020 [0.158] N=2,014	0.036 [0.173] N=1,813	0.120 [0.203] N=1,569	0.074 [0.229] N=1,296	-0.123 [0.263] N=980
BMI-for-age z score	0.123 [0.208] N=2,014	0.133 [0.229] N=1,808	0.254 [0.268] N=1,561	0.249 [0.317] N=1,289	0.022 [0.393] N=970
Stunted, HAZ below -2	-0.001 [0.039] N=2,046	-0.031 [0.044] N=1,835	-0.006 [0.048] N=1,585	0.018 [0.058] N=1,313	-0.016 [0.069] N=992
Underweight, WAZ below -2, through 120 months	0.018 [0.021] N=2,014	0.008 [0.022] N=1,813	-0.009 [0.024] N=1,569	-0.006 [0.029] N=1,296	0.011 [0.040] N=980
Thinness, BMIZ below -2	0.001 [0.030] N=2,014	-0.003 [0.032] N=1,808	0.009 [0.037] N=1,561	-0.013 [0.040] N=1,289	-0.003 [0.052] N=970
Overweight including obesity, BMIZ above +1	0.043 [0.060] N=2,014	0.032 [0.066] N=1,808	0.077 [0.076] N=1,561	0.036 [0.092] N=1,289	-0.033 [0.113] N=970
Obesity, BMIZ above +2	-0.006 [0.046] N=2,014	-0.001 [0.051] N=1,808	0.039 [0.060] N=1,561	0.021 [0.076] N=1,289	-0.123 [0.091] N=970

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.151	0.119 [0.136] N=2,273	0.191 [0.144] N=2,073	0.179 [0.150] N=1,834	0.081 [0.163] N=1,575	0.073 [0.185] N=1,279
Weight-for-age z score, through 120 months	0.404	0.041 [0.109] N=2,242	0.073 [0.113] N=2,052	0.135 [0.124] N=1,818	0.112 [0.128] N=1,558	0.095 [0.139] N=1,266
BMI-for-age z score	0.597	0.158 [0.143] N=2,240	0.184 [0.152] N=2,044	0.242 [0.164] N=1,808	0.220 [0.175] N=1,550	0.185 [0.200] N=1,256
Stunted, HAZ below -2	0.160 P=351	-0.001 [0.029] N=2,273	-0.019 [0.032] N=2,073	-0.002 [0.033] N=1,834	0.013 [0.036] N=1,575	-0.000 [0.040] N=1,279
Underweight, WAZ below -2, through 120 months	0.039 P=87	0.007 [0.015] N=2,242	0.001 [0.016] N=2,052	-0.006 [0.016] N=1,818	0.002 [0.018] N=1,558	0.006 [0.021] N=1,266
Thinness, BMIZ below -2	0.080 P=181	-0.013 [0.020] N=2,240	-0.018 [0.020] N=2,044	-0.009 [0.021] N=1,808	-0.015 [0.022] N=1,550	-0.015 [0.025] N=1,256
Overweight including obesity, BMIZ above +1	0.421 P=936	0.016 [0.043] N=2,240	0.013 [0.045] N=2,044	0.034 [0.048] N=1,808	0.013 [0.052] N=1,550	-0.007 [0.060] N=1,256
Obesity, BMIZ above +2	0.217 P=477	0.003 [0.034] N=2,240	0.009 [0.036] N=2,044	0.033 [0.039] N=1,808	0.022 [0.044] N=1,550	-0.011 [0.050] N=1,256

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.153 [0.189] N=1,925	0.294 [0.207] N=1,725	0.299 [0.220] N=1,486	0.121 [0.262] N=1,227	0.169 [0.325] N=931
Weight-for-age z score, through 120 months	-0.040 [0.171] N=1,894	0.004 [0.187] N=1,704	0.136 [0.220] N=1,470	0.077 [0.251] N=1,210	0.004 [0.297] N=918
BMI-for-age z score	0.097 [0.215] N=1,895	0.124 [0.239] N=1,699	0.266 [0.274] N=1,463	0.219 [0.322] N=1,205	0.059 [0.414] N=911
Stunted, HAZ below -2	-0.006 [0.040] N=1,925	-0.038 [0.046] N=1,725	-0.012 [0.048] N=1,486	0.021 [0.059] N=1,227	-0.028 [0.068] N=931
Underweight, WAZ below -2, through 120 months	0.015 [0.022] N=1,894	0.005 [0.024] N=1,704	-0.011 [0.026] N=1,470	-0.005 [0.031] N=1,210	0.002 [0.044] N=918
Thinness, BMIZ below -2	-0.003 [0.031] N=1,895	-0.014 [0.033] N=1,699	0.002 [0.039] N=1,463	-0.011 [0.044] N=1,205	-0.011 [0.060] N=911
Overweight including obesity, BMIZ above +1	0.024 [0.062] N=1,895	0.015 [0.068] N=1,699	0.066 [0.077] N=1,463	0.026 [0.092] N=1,205	-0.025 [0.118] N=911
Obesity, BMIZ above +2	-0.026 [0.048] N=1,895	-0.018 [0.053] N=1,699	0.026 [0.061] N=1,463	-0.002 [0.078] N=1,205	-0.141 [0.093] N=911

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.151	0.116 [0.137] N=2,273	0.187 [0.145] N=2,073	0.174 [0.152] N=1,834	0.078 [0.165] N=1,575	0.070 [0.188] N=1,279
Weight-for-age z score, through 120 months	0.404	0.044 [0.109] N=2,242	0.076 [0.113] N=2,052	0.141 [0.125] N=1,818	0.113 [0.128] N=1,558	0.097 [0.139] N=1,266
BMI-for-age z score	0.597	0.167 [0.146] N=2,240	0.192 [0.154] N=2,044	0.258 [0.166] N=1,808	0.232 [0.178] N=1,550	0.209 [0.201] N=1,256
Stunted, HAZ below -2	0.160 P=351	-0.002 [0.029] N=2,273	-0.019 [0.031] N=2,073	-0.002 [0.033] N=1,834	0.014 [0.036] N=1,575	-0.001 [0.040] N=1,279
Underweight, WAZ below -2, through 120 months	0.039 P=87	0.007 [0.015] N=2,242	0.001 [0.016] N=2,052	-0.005 [0.016] N=1,818	0.002 [0.018] N=1,558	0.005 [0.020] N=1,266
Thinness, BMIZ below -2	0.080 P=181	-0.012 [0.020] N=2,240	-0.017 [0.020] N=2,044	-0.009 [0.021] N=1,808	-0.015 [0.022] N=1,550	-0.017 [0.025] N=1,256
Overweight including obesity, BMIZ above +1	0.421 P=936	0.018 [0.043] N=2,240	0.015 [0.045] N=2,044	0.038 [0.049] N=1,808	0.018 [0.053] N=1,550	-0.000 [0.060] N=1,256
Obesity, BMIZ above +2	0.217 P=477	0.005 [0.035] N=2,240	0.011 [0.036] N=2,044	0.036 [0.040] N=1,808	0.025 [0.044] N=1,550	-0.007 [0.050] N=1,256

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.142 [0.189] N=1,925	0.280 [0.206] N=1,725	0.282 [0.219] N=1,486	0.087 [0.256] N=1,227	0.145 [0.315] N=931
Weight-for-age z score, through 120 months	-0.045 [0.171] N=1,894	-0.003 [0.187] N=1,704	0.130 [0.221] N=1,470	0.056 [0.251] N=1,210	-0.021 [0.297] N=918
BMI-for-age z score	0.094 [0.214] N=1,895	0.120 [0.239] N=1,699	0.265 [0.273] N=1,463	0.205 [0.322] N=1,205	0.047 [0.408] N=911
Stunted, HAZ below -2	-0.003 [0.040] N=1,925	-0.035 [0.045] N=1,725	-0.008 [0.048] N=1,486	0.031 [0.058] N=1,227	-0.019 [0.068] N=931
Underweight, WAZ below -2, through 120 months	0.015 [0.022] N=1,894	0.005 [0.024] N=1,704	-0.010 [0.026] N=1,470	-0.005 [0.031] N=1,210	0.001 [0.044] N=918
Thinness, BMIZ below -2	-0.002 [0.031] N=1,895	-0.013 [0.034] N=1,699	0.003 [0.039] N=1,463	-0.009 [0.043] N=1,205	-0.012 [0.059] N=911
Overweight including obesity, BMIZ above +1	0.024 [0.062] N=1,895	0.014 [0.068] N=1,699	0.066 [0.077] N=1,463	0.026 [0.092] N=1,205	-0.030 [0.116] N=911
Obesity, BMIZ above +2	-0.025 [0.048] N=1,895	-0.017 [0.053] N=1,699	0.027 [0.061] N=1,463	-0.003 [0.078] N=1,205	-0.142 [0.092] N=911

Notes: Cells report the reform coefficient, clustered standard error in brackets, and N; bold stars indicate *** p<0.01, ** p<0.05, * p<0.10. Five columns use 8- to 4-year parental birth-month bandwidths. The 12-month donut omits months -12 through +11. All specifications include exact child-month fixed effects, split linear trends, and FAKTOR_FERT survey weights. Specification 2 adds female; specifications 3 and 4 separately add household and per-person income groups; specifications 5 and 6 separately add aid count and any aid. WHO 2007 HAZ and BMIZ are valid at months 61-228; WAZ ends at month 120. Implausible z scores are excluded. P is the positive-case count in the eight-year no-donut sample. NA indicates zero standard error. School-level and measurement selection affect interpretation.